

Key Points:

- There is significant synchronization and periodicity between phytoplankton blooms and ENSO in the Western North Pacific
- ENSO drives phytoplankton blooms in the Western North Pacific show different patterns
- Phytoplankton blooms in the Western North Pacific strengthen with increasing Multivariate ENSO Index on the large spatiotemporal scale

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Teleconnection Between Coastal Phytoplankton Blooms Phenomenon in Western North Pacific and El Niño–Southern Oscillation by Time-Frequency Analysis

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Abstract Phytoplankton blooms in coastal marine environments are regarded as a double-edged sword, offering benefits to coastal ecosystems while also posing significant environmental challenges. The occurrence and severity of these blooms are influenced by environmental factors, with climate oscillations playing a crucial role. In this study, we employed two time-frequency analysis methods to investigate the teleconnection between phytoplankton blooms (denoted by bloom intensity, BI) and El Niño–Southern Oscillation (ENSO) from 2003 to 2020 in eight large marine ecosystems (LMEs) of Western North Pacific which are particularly sensitive to the impacts of climate oscillations and reflect well the impacts on marine ecosystems at different latitudes. These analytical methods unveiled these time series exhibited similar periodicity and synchronization patterns, which indicated the teleconnection between ENSO and phytoplankton blooms in the Western North Pacific. The Hilbert-Huang Transformation classified the eight LMEs into two categories based on their similarity to BI and MEI in the time-frequency domain. Specifically, one category is similar to MEI with energy concentrated at low frequencies, while the other category exhibits significant dissimilarity with dispersed energy. Wavelet analysis uncovered a robust coherence between the ENSO and BI across all LMEs, with a 57-month periodicity. Furthermore, phytoplankton blooms in the Western North Pacific Ocean strengthen with increasing Multivariate ENSO Index on the large spatiotemporal scale due to a decrease in sea surface temperatures. The findings will be useful as a reference for improving understanding of the teleconnection between coastal phytoplankton blooms and climate oscillations in other coastal zones.

Plain Language Summary It is accepted that phytoplankton biomass in the Western North Pacific exhibits significantly higher levels during La Niña years compared to El Niño years as a result of rising sea temperature, thereby stimulating primary productivity. However, this ignores the complex influence of climate oscillations in the context of the overall trend of global warming. In this study, long-term satellite remote sensing data from eight Western North Pacific large marine ecosystems (2003–2020) showed that phytoplankton blooms in the Western North Pacific Ocean strengthen with increasing Multivariate ENSO Index on large spatiotemporal scale, that is, the interannual temporal scale and marine spatial scale, primarily due to a decrease in sea surface temperatures. Furthermore, the impact of ENSO on phytoplankton in Western North Pacific can be categorized into two types: one characterized by the direct effect of ocean currents and the other characterized by coupling of atmosphere and ocean.

1. Introduction

Coastal ecosystems have attracted intense attention and interest because of their sensitivity to climate and their economic and social importance to humans (Dai et al., 2023; Temmerman et al., 2013; Waltham et al., 2020). Coastal waters are highly dynamic ecosystems that may be particularly sensitive to the impacts of climate oscillations due to the tight associations between atmospheric forcing, nutrient inputs, and propagule transport in marine ecosystems (Carstensen et al., 2015). Phytoplankton is the link between higher trophic level organisms and the climate by fixing carbon at the base of the food chain and supporting fisheries and ecosystems worldwide (Defriez et al., 2016), which is important for energy flow, material circulation, and information transfer in marine ecosystems (Behrenfeld et al., 2001; Rousseaux & Gregg, 2013; Sun et al., 2022; Amin et al., 2015). Therefore, studying phytoplankton trends becomes essential to understand the response between climate and coastal zone ecosystems.

In general, phytoplankton growth is believed to be influenced by several factors, such as increasing irradiance, temperature, and nutrient inputs (Colijn & Cadée, 2003; Glé et al., 2007; Tian et al., 2011). Therefore, research on phytoplankton focuses more on some direct influences, such as sea surface temperature (SST), mixed layer depth (MLD) and nutrients, in which the impacts of human activities in the coastal zone are also the focus of research, and the attention on climate events also focuses on small scale events, such as typhoons (Li et al., 2022; Shibano et al., 2011; Lin et al., 2003), eddies (Chang et al., 2017; Vidya et al., 2021; Waga et al., 2019) and upwelling (Le et al., 2022; Lu et al., 2015; Huang et al., 2011), and insufficient attention is paid to large-scale external forcing, such as the macroscopic impacts of climate oscillations.

Climate change is recognized as a potential catalyst for the expansion of phytoplankton blooms, affecting the physical-chemical environment and biological processes in water bodies (Lim et al., 2022; Paerl et al., 2011; Paerl & Huisman, 2008; Paerl & Paul, 2012). Temperature and rainfall changes, influenced by climate change, play a significant role in simulating the blooms (Xiao et al., 2019). El Niño–Southern Oscillation (ENSO) are episodic departures from the expected SST in the equatorial Pacific Ocean, which can affect weather patterns around the world by influencing high and low pressure systems, winds, and precipitation (Cai et al., 2021; Power et al., 2021). Researchers reported that precipitation and temperature in Western North Pacific might have an ENSO related response, such as the Western North Pacific anomalous anticyclone (WNPAC), Pacific–Japan and Silk Road teleconnections (Li et al., 2017; Zhang & Zhou, 2015). Furthermore, extreme climate events, such as extreme temperature and marine heatwaves, have also been shown to be closely related to ENSO (Li et al., 2023; Joh & Di Lorenzo, 2017; Guo et al., 2022). Given that the ENSO can influence temperature, precipitation, and extreme climate events, the ENSO would also drive the phytoplankton as a result of interlocking.

Marine phytoplankton are naturally believed and already proven to be influenced by climate oscillations. For instance, although strong correlations between marine phytoplankton and modes of climate variability (i.e., North Atlantic Oscillation, El Niño–Southern Oscillation) were reported for several lakes in coastal areas, much less is known for large scales in the marine as a unit, comparing to small and medium scales (i.e., bays, estuaries) (Santos et al., 2021, 2022). Furthermore, on large level, the ultimate drivers of climate oscillations on phytoplankton are not yet understood, and the consequences of the ecosystem remain uncertain (Menge et al., 2009). Given the current limitations in understanding coastal ecosystems, it becomes imperative to explore these issues from a fresh perspective. In this sense, data-driven statistical methods emerge as a valuable approach to compensate for the lack of knowledge about physical mechanisms and help us to comprehend phytoplankton in response to climate change across large spatial and temporal scales (Liu et al., 2023).

Recently, many data-driven statistical methods have been developed to analyze nonstationary and nonlinear time series data (Somu et al., 2021; Watorek et al., 2021; Tiyasha et al., 2020). Hilbert-Huang transformation (HHT), consisting of empirical mode decomposition (EMD) and the Hilbert transformation (HT), is an advanced time-frequency analysis technique that addresses the shortcomings of the Fourier transformation, which has a priori basis rather than adaptive and is only reliable for linear and stationary processes (Huang et al., 1998; Huang & Wu, 2008). Global information, that is, the intrinsic mode of variables, provides a chance for gaining better insights into their dynamic pattern analyzed by Hilbert spectra; besides, local information, that is, local frequency, amplitude and phase, etc., would allow a better understanding of time-frequency variation characteristics for the inputs (Hu et al., 2016; Palacz et al., 2011). It has been applied to studies related to phytoplankton blooms and used to establish the relationship between phytoplankton blooms and climate changes. For example, Xiao et al. employed HHT to explore the relationship between phytoplankton in American lakes and ENSO (Xiao et al., 2019) and Palacz et al. employed ensemble empirical mode decomposition (EEMD) to explore seasonal and inter-annual changes in the surface chlorophyll of the South China Sea (Palacz et al., 2011). The wavelet transform (WT) is well known to hold considerable promise in the analysis of non-stationary and stationary variations of time series data. As an applied form of wavelet transform, wavelet coherence analysis, the inter-relationship between two correlated variables can be represented in the time-frequency domain. So far it has been widely applied in studies of climate changes (Cheng et al., 2021; Jerin et al., 2021), as well as the trend of phytoplankton variation (Carey et al., 2016; Hieronymus et al., 2018). Up-to-date data-driven statistical methods would provide us with a new opportunity to explore the relationship between phytoplankton and the complex signals of climate oscillations and thus bring new insights into how extreme water quality and coastal ecosystem productivity are related to climate changes.

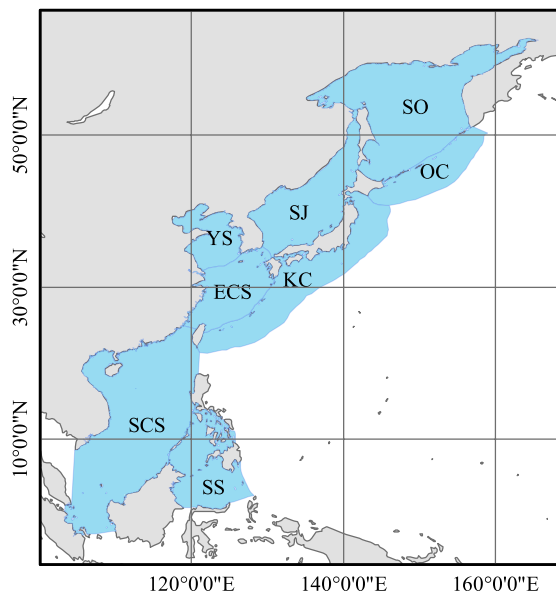


Figure 1. Sketches of the eight Western North Pacific LMEs. SCS = South China Sea, SS = Sulu-Celebes Sea, ECS = East China Sea, YS = Yellow Sea, KC = Kuroshio Current, SJ = Sea of Japan, SO = Sea of Okhotsk, OC = Oyashio Current.

In view of the above considerations, to improve our understanding of the teleconnection between coastal phytoplankton blooms and climate oscillations, this study focused on marine scale and applied different time-frequency analysis methods to reveal the periodicity and synchronization of phytoplankton blooms and ENSO. The primary objectives of this study were as follows: (a) the intuitive association between phytoplankton blooms and ENSO within eight large marine ecosystems (LMEs) in the Western North Pacific Ocean based on marine scales; (b) the intrinsic periods of phytoplankton and ENSO in a long term in the eight LMEs in the Western North Pacific Ocean, employing HHT in tandem with its corresponding spectrum, as well as the wavelet correlation analyses; (c) the corresponding spatial variability between phytoplankton blooms and climate oscillations in different LMEs. The findings will be useful as a reference for improving the understanding of the teleconnection between coastal phytoplankton blooms and climate oscillations in other coastal zones.

2. Materials and Methods

2.1. Study Area and Time Series Data Acquisition

2.1.1. Study Area

Eight LMEs within the Western North Pacific were studied, including the South China Sea (SCS), Sulu-Celebes Sea (SS), East China Sea (ECS), Yellow Sea (YS), Kuroshio Current (KC), Sea of Japan (SJ), Sea of Okhotsk (SO), Oyashio Current (OC) (as shown in Figure 1). LMEs encompass global

coastal oceans and outer edges of coastal currents areas, which are defined by various distinct features of the oceans, including hydrology, productivity, bathymetry, and trophically dependent populations. Of the 66 LMEs identified globally, we chose the Western North Pacific regions and examined 8 LMEs. The latitude range of these eight LMEs includes 3° S–63° N, which can well reflect the latitudinal trends. The boundaries of LMEs were obtained from <https://www.sciencebase.gov/catalog/item/55c77722e4b08400b1fd8244>.

2.1.2. Phytoplankton Blooms

In general, Chlorophyll *a* (CHL*a*) concentrations is often used to indicate phytoplankton biomass since the proof-of-concept Coastal Zone Color Scanner (1978–1986) era, which has been provided as a standard product by NASA (Gordon et al., 1980; Moore & Abbott, 2000). The default algorithm for retrieving CHL*a* products rely on the high absorption of CHL*a* in the blue band (Hu et al., 2019; Hu et al., 2012). However, this can lead to increased uncertainty in coastal regions because the absorption of CHL*a* in the blue band may be obscured by high levels of suspended sediments and/or colored dissolved organic matter (Hu & Feng, 2017). To mitigate this issue, numerous spectral indices, such as the normalized fluorescence line height (nFLH) (Hu et al., 2005), red tide index (RI) (Ahn & Shanmugam, 2006), algal bloom index (ABI) (Hu & Feng, 2017), red-blue difference (RBD) (Amin et al., 2009), have been developed that focus on identifying phytoplankton blooms rather than quantifying their biomass using CHL*a*. To conduct a large-scale analysis of phytoplankton blooms, this paper uses the dataset produced by the CIE - fluorescence algorithm proposed by Hou et al. (2022) and Dai et al. (2023), which can more accurately identify coastal phytoplankton blooms that are affected by green, yellow, brown, or even red phytoplankton by jointly inverting the remotely sensed reflectance of three visible bands (red, green, and blue) associated with these phytoplankton blooms. Its validation was conducted using the Harmful Algae Event Database (HAEDAT) (<http://haedat.iode.org>). In the generation of this dataset, diverse threshold conditions were implemented across various regions. This approach effectively addresses the challenge posed by significant variations in phytoplankton biomass below the first optical depth observed in distinct regions (Schulien et al., 2017). Consequently, this dataset, characterized by the presence or absence of blooms, provides a more comprehensive reflection of the trends of phytoplankton changes across different regions. The phytoplankton blooms dataset (<https://doi.org/10.5281/zenodo.7359262>) covers a period from 1 January 2003, to 31 December 2020, with a spatial resolution of 0.01°. The pixel values within the data are classified into four categories: 0, 99, 100, and 255, representing no blooms, blooms, nodata, and areas masked out by land, clouds, or other factors, respectively.

We defined bloom intensity (BI, dimensionless) to represent the density of phytoplankton blooms for a month integrating the bloom count and bloom-affected areas within $0.25^\circ \times 0.25^\circ$ grid cells within that month, while excluding the effect of nodata and masked areas in remote sensing interpretation. The equation for BI is as follows:

$$BI = \frac{1}{n} \sum_{i=1}^n \frac{m * B_i}{B_i + N_i}$$

where B_i represents the enumerated blooms count for each $0.01^\circ \times 0.01^\circ$ resolution pixel in one day within one $0.25^\circ \times 0.25^\circ$ grid cell, while N_i represents the enumerated noblooms count for each $0.01^\circ \times 0.01^\circ$ resolution pixel in one day within the same $0.25^\circ \times 0.25^\circ$ grid cell, m is the number of original resolution pixels contained in the resampled grid cells, here the value is 125, and n is the number of days in a month.

2.1.3. Multivariate ENSO Index

The Multivariate ENSO Index (MEI), which combines both oceanic and atmospheric variables, facilitates in a single index an assessment of ENSO and is regarded as a better and synthesized proxy of ENSO that causes global climate variability on interannual time scales (Wolter & Timlin, 2011). It especially gives real-time indications of ENSO intensity, and through historical analysis provides a context for a meaningful comparative study of evolving conditions. It was used to examine the connections between El Niño–Southern Oscillation activities and marine phytoplankton blooms. Monthly MEI values for the same period of January 2003 to December 2020, as the index of ENSO in this study, were downloaded from the National Oceanic and Atmospheric Administration (NOAA) website (<https://psl.noaa.gov/enso/mei/>) (as shown in Figure 2). In general, elevated positive values of the MEI indicate the presence of El Niño phase, characterizing a warm ENSO phase. Conversely, high negative MEI values indicate the occurrence of significant La Niña phase, signifying a cold ENSO phase.

2.1.4. Marine Environmental Factors

SST and MLD are important marine environmental factors affecting phytoplankton growth (Carvalho et al., 2017; Dai et al., 2023; Trombetta et al., 2019), which can be seen as a bridge between climate oscillations and phytoplankton blooms. In this study, we mainly aim to investigate how climate oscillations affect phytoplankton by driving changes in SST and MLD and to establish the teleconnection between coastal phytoplankton blooms and ENSO. The SST data was downloaded from Climate Data Store (CDS) at DOI:10.24381/cds.adbb2d47. The horizontal resolution is $0.1^\circ \times 0.1^\circ$ and the temporal resolution is 1 hr. The MLD was downloaded from CDS at DOI:10.24381/cds.67e8eeb7. The MLD is defined as the depth of the ocean where the average sea water density exceeds the near surface density plus 0.03 kg/m^3 . The horizontal resolution is $0.25^\circ \times 0.25^\circ$ and the temporal resolution is 1 month.

2.2. Time Series Analysis Methods

2.2.1. Hilbert-Huang Transformation

In this study, the monthly BI time series for the eight LMEs in Western North Pacific and the monthly MEI time series were analyzed by the HHT individually. Normally, HHT consists of EMD and HT (Huang et al., 1998; Huang & Wu, 2008). EMD divides the original time series into several independent time series known as intrinsic mode functions (IMFs):

$$x(t) = \sum_{j=1}^n c_j(t) + r_n$$

where $x(t)$ represents the original time series, $c_j(t)$ represents the IMFs, r_n represents the residue of $x(t)$, after the n number of IMFs are extracted. However, the EMD process can sometimes cause mode mixing problems (Wu & Huang, 2009). To address this issue, this study used EEMD based on ensemble ideas, which is a modified version of EMD that can effectively solve the mode mixing dilemma. EEMD involves adding white noise to the original signal and performing EMD on each of the noisy versions. The resulting IMFs are then averaged to obtain the final IMF.

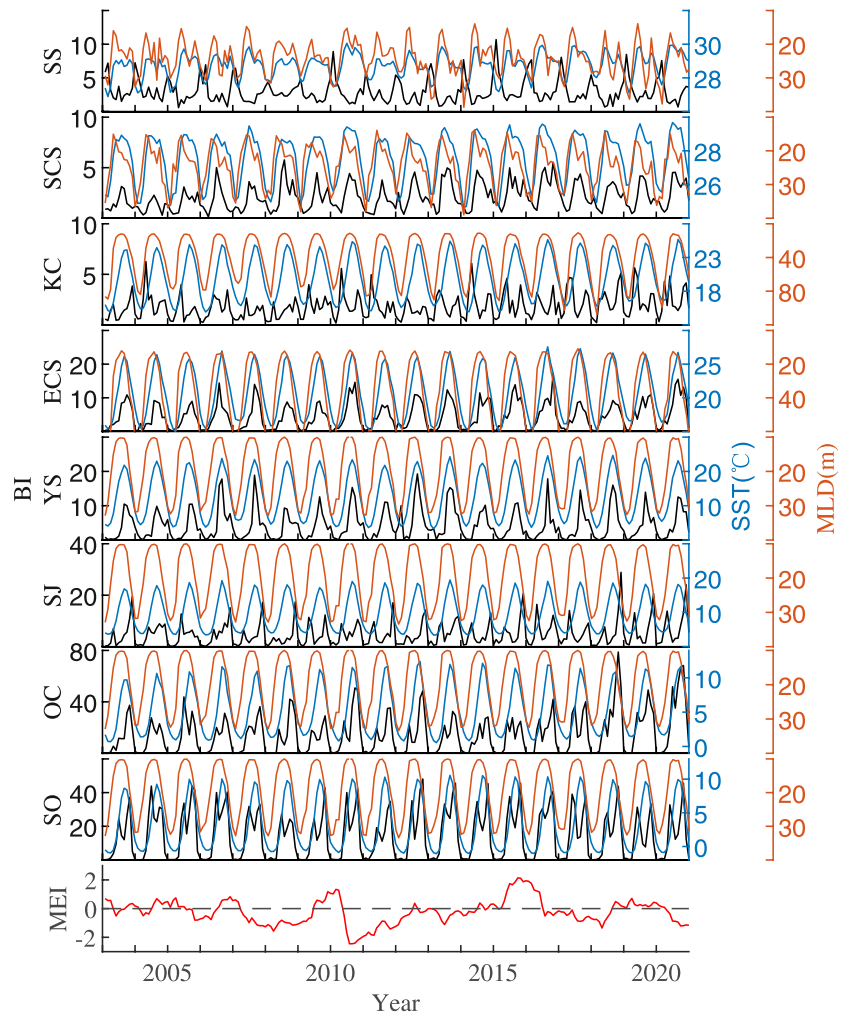


Figure 2. Original time series of BI, SST, and MLD in the eight Western North Pacific LMEs and MEI from January 2003 to December 2020. The black line represents BI; the cyan line represents SST; the magenta line represents MLD; the red line represents MEI.

HT transforms each IMF derived from the EEMD into an analytical function that is used to construct the Hilbert spectrum, which represents the time-frequency-amplitude (energy/variance) characteristics of the original time series in the time-frequency domain. The original time series $x(t)$ can be expressed as the real part, Re, of the sum of the data in terms of time, frequency, and amplitude as:

$$x(t) = \text{Re} \left[\sum_{j=1}^n a_j(t) e^{i \int \omega_j(t) dt} \right]$$

where $a(t)$ and $\omega(t)$ represent the instantaneous amplitude and frequency functions for each component, respectively. With the Hilbert spectrum, the marginal spectrum $h(\omega)$ is computed as a cumulative amplitude distribution across frequencies, it can be calculated as follows:

$$h(\omega) = \int_0^T H(\omega, t) dt$$

The Hilbert marginal spectrum expresses the total amplitude contribution of each frequency value. This function represents the cumulative amplitude over the entire range of data in a probabilistic sense. In order to quantitatively

describe the marginal spectrum, Frequency Entropy is used to describe its distributional uncertainty or complexity. Shannon Entropy is widely used and is calculated as follows:

$$H(f) = - \sum_{i=1}^N P(f_i) \log_2(P(f_i))$$

where N represents the number of discrete points of the marginal spectrum, $P(f_i)$ represents the normalized amplitude of the frequency component f_i . Higher Frequency Entropy indicates a more uniform distribution of energy in the spectrum, while lower Frequency Entropy indicates a concentration of energy at a few frequencies.

2.2.2. Wavelet Analysis

Wavelet analysis was performed to explore the interresponse relationship between the monthly time series of BI for each of the eight LMEs and the monthly time series of MEI. The specific method chosen in this study is wavelet correlation analysis (Cazelles et al., 2008; Grinsted et al., 2004), which uses the Morlet wavelet function as the mother function. The time series can be decomposed dependent on time and scale by continuous wavelet transform (CWT):

$$W_t^x(s) = \sqrt{\frac{\delta t}{s}} \sum_{t'=0}^{t-1} x_{t'} \psi_0\left(\frac{(t' - t)\delta t}{s}\right)$$

where x_t is the analyzed time series that is recorded in a discrete time sequence (t) with a uniform interval of δt ; and s represents the scaling factor of the mother wavelet. The mother wavelet depends on a time variable, $\eta = s.t$, which normalizes the CWT to have unit energy. This normalization allows all scales to be directly comparable to each other. The Morlet wavelet can be expressed as:

$$\psi_0(\eta) = \pi^{-1/4} e^{i\omega_0\eta} e^{-\eta^2/2}$$

where $\omega_0 = 6$ is the dimensionless frequency for the Morlet wavelet due to the good balance between time and frequency localizations. η is dimensionless time.

After the two transformed series were cross-wavelet transformed to get the cross wavelet spectrum by cross wavelet transform (XWT). The XWT of two time series x_n and y_n can be defined as:

$$W^{XY} = W^X W^{Y*}$$

where $*$ denotes complex conjugation, W^X and W^Y represent the CWT of two time series x_n and y_n , respectively.

Additionally, the coherency wavelet values were calculated in the time-frequency domain, using the individual wavelet spectrum to normalize the cross wavelet spectrum. The wavelet coherence of two time series can be defined as:

$$R_t^2(s) = \frac{|S(s^{-1} W_t^{xy}(s))|^2}{S(s^{-1} |W_t^x(s)|^2) \cdot S(s^{-1} |W_t^y(s)|^2)}$$

where $W_t^x(s)$ and $W_t^y(s)$ are the CWT of time series X and Y , respectively. $W_t^{xy}(s)$ is the cross wavelet transform and S is the smoothing operator.

Furthermore, we use phase analysis to characterize the correlation between time series. The phase difference provided information on the sign of the relationship (i.e., in phase or out of phase). The phase was computed based on the imaginary and the real part of the wavelet transform of the time series, and the phase difference was based on the imaginary and the real part of the wavelet co-spectrum between two time series.

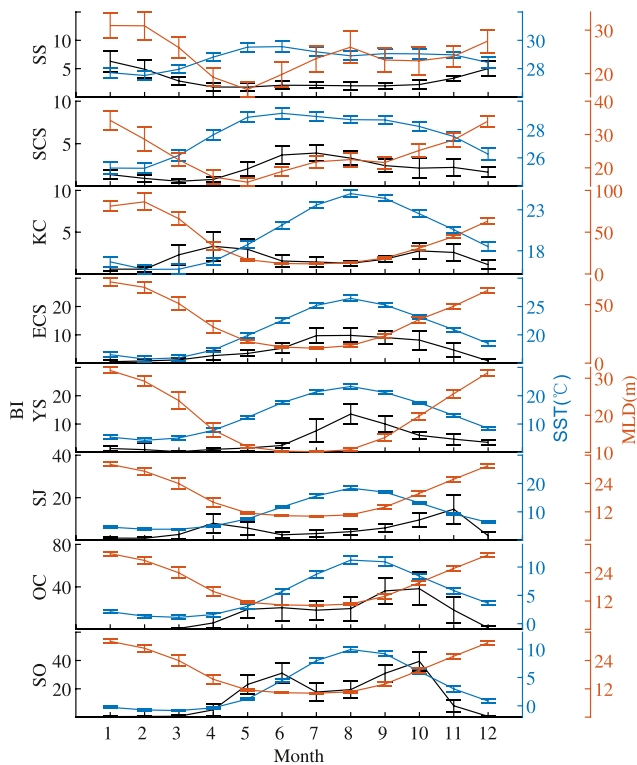


Figure 3. Seasonal cycle of BI, SST, and MLD in the eight Western North Pacific LMEs from January 2003 to December 2020. The black line represents BI; the cyan line represents SST; the magenta line represents MLD; Vertical error bar represents 95% confidence level.

calculated as 0.6662, -0.2319 , -0.2810 , -0.6794 , -0.4201 , -0.0998 , -0.4899 , -0.6532 , respectively. These results suggested a negative correlation between BI and MLD across LMEs except for the SS, where a positive correlation is evident. This also suggested the opposite correlation between SST/MLD and BI. SJ and KC are significantly affected by warm currents (Kuroshio Current and Tsushima Warm Current), and temperature is no longer a constraint on phytoplankton growth, so BI has a lower correlation with SST. Additionally, MLD in warm currents is also relatively more stable, with less seasonal variations, so therefore BI has a lower correlation with MLD.

3.2. Multi-Timescale Correlation

3.2.1. Seasonal Cycle

The seasonal cycle provides a better answer to the synchronization between the BI and the marine environmental factors. The seasonal cycle of the BI revealed a discernible response to its seasonal regularity and its association with the SST and MLD, shown as Figure 3. Excluding SS, SJ, and SO, the BI within the remaining five LMEs all demonstrated a robust positive correlation with SST. The correlation coefficients for SCS, KC, ECS, YS, and OC were calculated as 0.7716, 0.8415, 0.8655, 0.9669, and 0.8083, respectively. The peaks of BI in SCS and the OC lagged relative to the peak of SST, the peak of BI in the YS coincided with the peak of SST, while the peaks of BI in ECS and the KC advanced relative to the peak of SST. This result suggests that in SCS and OC, the increase in temperature does not significantly constrain the outbreak of phytoplankton blooms even in summer, whereas in ECS and KC, the increase in temperature in summer has an inhibitory effect on the outbreak of phytoplankton blooms. Additionally, a significant negative correlation between BI and SST is observed in SS, with a correlation coefficient of -0.7913 . On the contrary, in SJ and SO, the BI showed a decrease with increasing SST during the summer months and an increase during the rest of the year. These two LMEs showed a clear spring bloom and an autumn bloom, thus constituting a bimodal phenomenon in the

2.2.3. Local Polynomial Regression Fitting

Local Polynomial Regression Fitting (LOESS) is a nonparametric method for local regression analysis (Gutierrez et al., 2003), and it was employed to fit the annual mean BI in the eight LMEs against MEI values. LOESS is a smoothing technique that uses a moving window to search for neighboring data points and assigns different weights to each data point based on its distance from the point being fitted. This allows for flexible and adaptive fitting of the data, without imposing strict assumptions about the underlying functional form of the relationship. The LOESS analysis was performed using R software, to explore the intuitive associations between the BI and MEI values in the different LMEs, as a complement to time-frequency analysis.

3. Results

3.1. Original Signals

To carry out the study of the phytoplankton blooms in the study areas, we spatially averaged the 216-month dataset by LMEs to obtain monthly BI data for these 8 LMEs based on the BI dataset obtained from the previous dataset and definitions. SST and MLD are important factors influencing phytoplankton blooms. SST and MLD were spatially averaged over 8 LMEs to obtain their time-varying signal. The above and the original MEI signals are shown in Figure 2. It is easy to see that the seasonal regularity of the BI, like SST and MLD, in all 8 LMEs, despite the range of BI (0–5 in SS to >40 in SO), which indicated the correlation between BI and SST/MLD. Specifically, the correlation coefficients between BI and SST for SS, SCS, KC, ECS, YS, SJ, OC, and SO were calculated as -0.6664 , 0.6618, 0.0187, 0.8540, 0.7751, 0.1579, 0.6780, 0.6789, respectively. These results suggested a positive correlation between SST and BI across LMEs except for SS, where a negative correlation is evident. The correlation coefficients between BI and MLD for SS, SCS, KC, ECS, YS, SJ, OC, and SO were

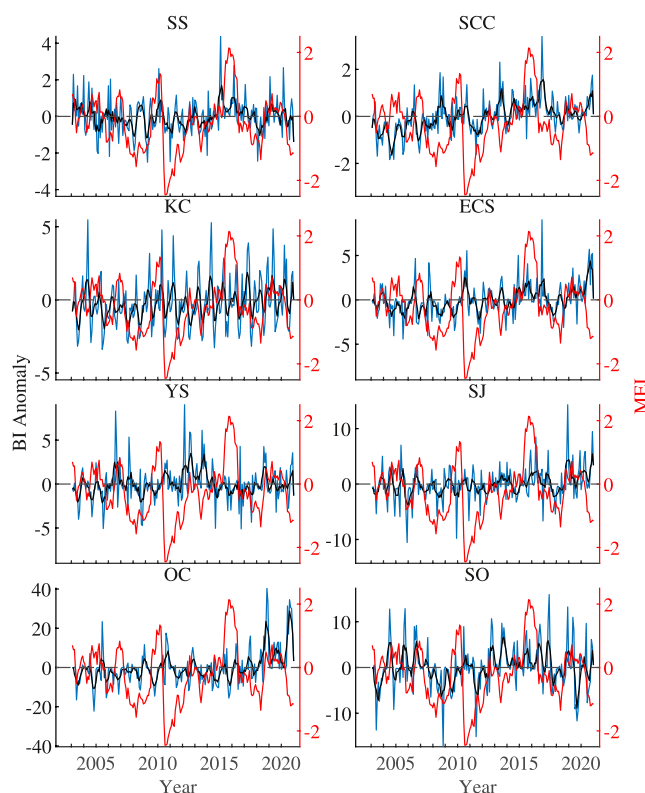


Figure 4. Anomaly of BI in the eight Western North Pacific LMEs and MEI from January 2003 to December 2020. The cyan line represents anomaly of BI; the black line represents smoothed anomaly of BI; the red line represents MEI.

seasonal cycle of the BI. The increase in SST in summer had a significant inhibitory effect on phytoplankton growth and reproduction, and thus a downward trend in the BI was observed. When it comes to MLD, the BI within the remaining five LMEs, excluding SS, SCS, YS, and SJ, all demonstrated a robust negative correlation with MLD. The correlation coefficients for KC, ECS, OC, and SO were calculated as -0.7653 , -0.7717 , -0.6038 , and -0.6989 , respectively. Additionally, a significant positive correlation between BI and MLD is observed in SS, with a correlation coefficient of 0.8246 .

In addition, because ENSO is an interannual climate variability, the seasonal cycle of BI can be removed to get the anomaly of BI, shown as Figure 4. The correlation coefficients between smoothed anomaly of BI and MEI for SS, YS, SJ, OC, and SO were calculated as 0.4917 , -0.1713 , 0.1943 , -0.1782 , and 0.1681 , respectively. And the correlations of those LMEs were significant ($P > 0.05$). Additionally, the correlation coefficients of SCS, KC, ECS were 0.0073 , 0.0986 , 0.0027 ($P \geq 0.05$). Among the eight LMEs, only the anomaly of BI in SS and MEI showed a strong positive correlation. Considering that this direct linear correlation is small in most LMEs, it makes it necessary to study the correlation and regression between MEI and BI from other perspectives.

3.2.2. Intuitive Association Between Annual Average BI and MEI

Considering that the main periods of BI in eight Western North Pacific LMEs were expressed in approximately 12 months, the annual average BI and MEI values were calculated for intuitive association analysis (Figure 5) by LOESS fitting. With the MEI value increasing, BI in eight Western North Pacific LMEs experienced a general trend of “increasing-decreasing-increasing-decreasing”. The inflection points located around MEI values of -0.8 and 0.8 , respectively. Especially, the annual average BI exhibits a smaller range of fluctuations during La Niña phase (when the MEI value is less than 0, the mean variance of the BI for the different LMEs is 2.68) compared to the El Niño phase (when the MEI value is greater than 0, the mean variance of the BI for the different LMEs is 5.09).

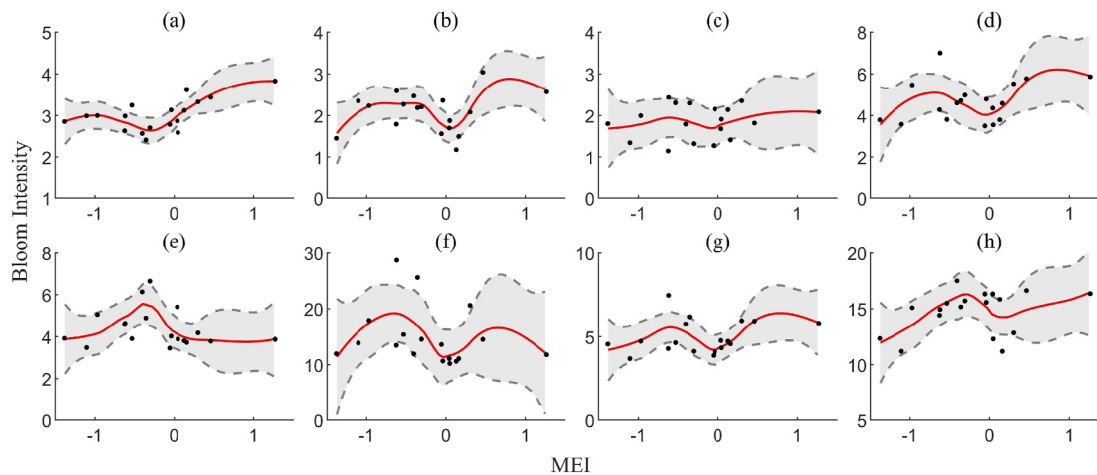


Figure 5. Intuitive associations between BI in eight Western North Pacific LMEs and MEI in annual scale. (a)–(h) represents the intuitive associations for SS, SCS, KC, ECS, YS, SJ, OC, and SO. The red lines on scatter plots indicate a smooth curve fitted by LOESS regression, and the shadow region shows the confidence interval at 0.05 level.

3.3. Time-Frequency Analysis

3.3.1. Ensemble Empirical Mode Decomposition

EEMD can decompose the BI time series and MEI time series into complex signals superimposed by signals with different periods, which contain multiple implied periods, reflecting the period evolution characteristics and fluctuation structure of the two phenomena, and the period-based coupling is conducive to the construction of the correlation between the BI and MEI. For each of the eight BI and the MEI timing signals, six IMFs mode components and a residue mode component were generated by the EEMD method of HHT (Figure 6). The oscillation periods of the BI and ENSO time series were calculated, with the statistical results of cross-comparison shown in Table 1. Similar patterns were observed for the nine time series. The frequency of time series decreased from IMF1 to IMF6, and residuals were mostly monotonic except for SO. All IMFs oscillated across the entire series. The average period of IMF1 was about 3 months, which reflects the seasonal cycle evolution characteristics and small fluctuation structure of BI and MEI, and the average period of IMF2 is about 6 months, which reflects the semiannual cycle evolution characteristics and fluctuation structure of BI and MEI, and the average period of IMF3 is about 12 months, which reflects the annual cycle evolution characteristics and fluctuation structure of BI and MEI. The average period of IMF4–6 is more than 24 months (i.e., 2 years), reflecting the interannual cycle evolution and volatility structure of BI and MEI. And the residue is monotonic over the study period (i.e., the period is greater than 18 years), reflecting the decadal/multi-decadal cyclical evolutionary characteristics and volatility structure of the BI and MEI. The cyclic behavior presented by the mode components, the EEMD decomposition method can well reflect the multi-timescale characteristics of phytoplankton bloom and ENSO, that is, seasonal scale, semiannual scale, annual scale, interannual scale, and decadal/multi-decadal scale characteristics.

The eight time series of BI exhibited a dominance of IMF 1–3 with significant variance contributions, and IMF1–3 showed approximate periods of 3, 6, and 12 months. The highest variance contribution of IMF3 was found in the decomposition results of SS, SCS, ECS, and YS, while the highest variance contribution of IMF1 was found in the decomposition results of KC, SJ, OC, and SO. The decomposition results of MEI were dominated by IMF3–6 with the largest variance contribution from IMF4. Although the IMF3, with a mean period of 16.07 months, is not the mode component with the highest variance contribution, it still exhibits a dominant mode with a significant variance contribution of 13.90%, and it can be regarded as the annual cyclical fluctuation mode of the MEI, that is, the ENSO, within a certain margin of error. For the BI time series within the eight LMEs, its annual cycle-based fluctuation pattern (IMF3) is dominant in the vast majority of the cases, and its average variance contribution can reach 27.47%. This similarity of the dominant cycles based on the annual cycle reflects the similarity of the phytoplankton bloom and ENSO in the eight LMEs in terms of the evolutionary characteristics and fluctuation structure to a certain extent.

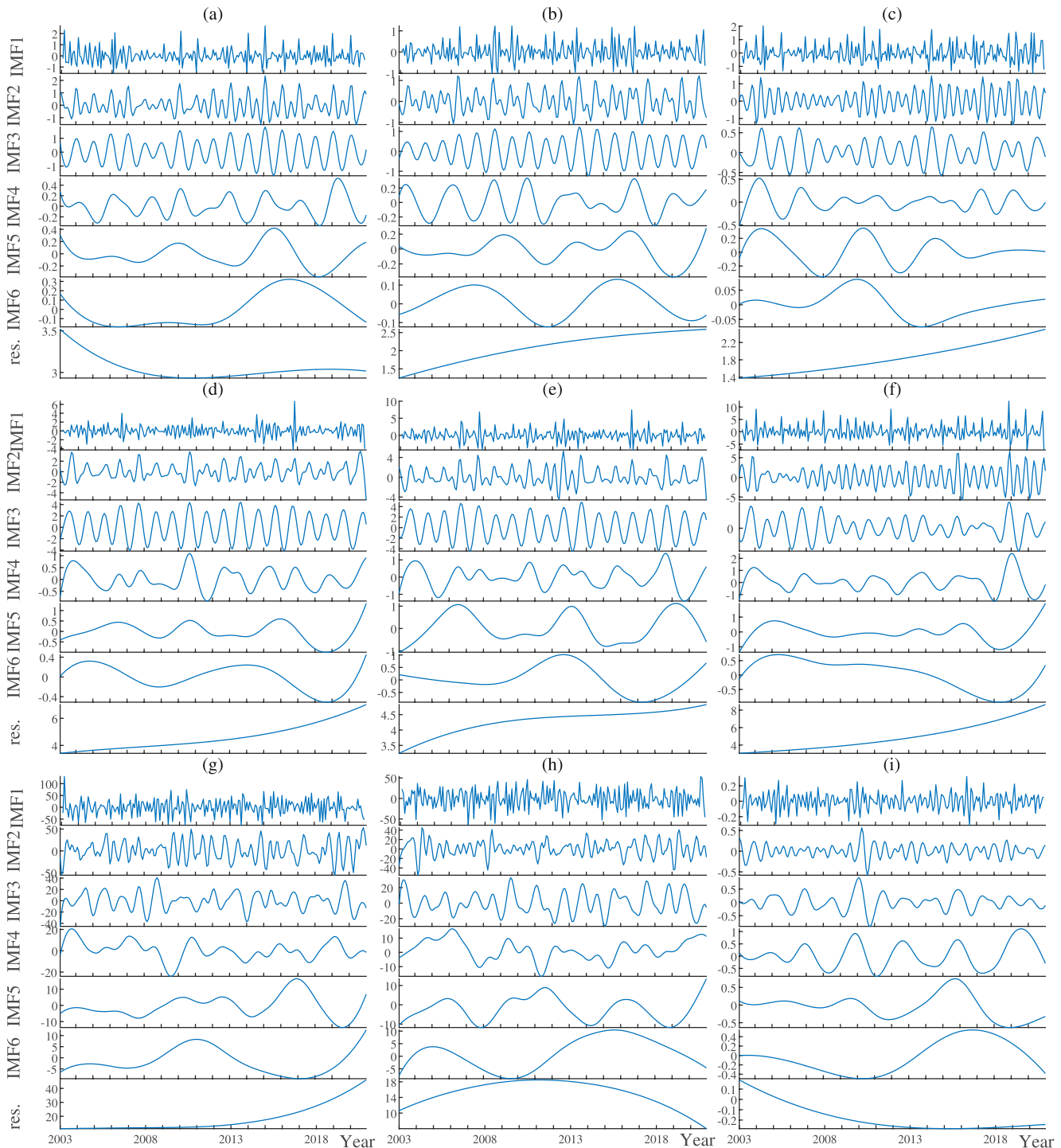


Figure 6. EEMD results of the BI in eight Western North Pacific LMEs - SS (a), SCS (b), KC (c), ECS (d), YS (e), SJ (f), OC (g) and SO (h), and MEI (i). IMF 1–6 represent the intrinsic mode function generated by EEMD, respectively; res. represents the residue after EEMD process.

3.3.2. Hilbert Spectra and Marginal Spectra

HT is able to achieve an understanding of the characteristics of time-frequency variations of signals by localizing local information such as frequency, amplitude and phase of the periodic signals decomposed by EEMD, thus discovering the correlation between different signals. Time-frequency-energy (amplitude) distribution of the BI

Table 1
Statistics of EEMD Results of the BI Eight Western North Pacific LMEs and MEI

LMEs	Model	Mean period (month)	Variance	LMEs	Model	Mean period (month)	Variance
SS	IMF1	2.94	25.14	SCS	IMF1	3.09	17.54
	IMF2	6.27	29.59		IMF2	6.21	26.15
	IMF3	12.24	39.10		IMF3	12.12	36.17
	IMF4	22.78	2.01		IMF4	26.40	2.85
	IMF5	54.00	1.73		IMF5	43.25	2.15
	IMF6	80.00	1.57		IMF6	101.33	0.57
KC	IMF1	2.74	40.63	ECS	IMF1	2.91	15.10
	IMF2	6.06	34.95		IMF2	7.05	19.90
	IMF3	12.18	7.93		IMF3	12.18	52.56
	IMF4	25.87	3.21		IMF4	20.21	1.70
	IMF5	60.67	3.20		IMF5	41.71	1.77
	IMF6	78.00	0.16		IMF6	110.67	0.56
YS	IMF1	3.65	21.94	SJ	IMF1	2.88	44.09
	IMF2	6.03	22.51		IMF2	6.06	29.93
	IMF3	12.11	46.66		IMF3	12.24	8.45
	IMF4	22.24	2.70		IMF4	23.18	2.81
	IMF5	38.25	3.05		IMF5	45.14	1.49
	IMF6	110.00	2.07		IMF6	104	1.31
OC	IMF1	2.74	52.92	SO	IMF1	2.81	47.25
	IMF2	5.81	24.37		IMF2	5.34	24.03
	IMF3	10.40	11.63		IMF3	12.41	17.25
	IMF4	20.50	3.44		IMF4	17.45	4.25
	IMF5	41.56	2.14		IMF5	44.29	3.19
	IMF6	98.67	1.43		IMF6	127	3.14
MEI	IMF1	3.31	1.90	MEI	IMF4	34.55	40.18
	IMF2	6.83	3.85		IMF5	60.00	19.18
	IMF3	16.07	13.90		IMF6	158	18.19

for the eight LMEs and MEI time series were further calculated as the Hilbert spectrum by HT (Figure 7). All nine time series were analyzed using the Hilbert spectrum, and their results can be categorized into three groups. The first group, consisting of SS, SCS, ECS, and YS, showed a stable and continuous instantaneous frequency distribution in the low frequency region (about 12 months per cycle or 0.08 cycles/month), with their energy concentrated in the high frequency region (about 3 months per cycle or 0.3 cycles/month). On the other hand, the second group, including SJ, OC, KC, and SO, exhibited no continuous and stable instantaneous frequency distribution, and their instantaneous frequencies were discrete and oscillating. Similarly, the energy for this group was also concentrated in the same high frequency region. Finally, for MEI, a continuous and stable instantaneous frequency distribution was found in the low frequency region (more than 33 months per cycle or less than 0.03 cycles/month), with the energy (variance) concentrated in this region.

As a cumulative amplitude distribution, the marginal spectrum would allow a better demonstration of the amplitude (energy/variance) characteristics of the time series (Figure 8). All of Hilbert marginal spectrum showed similar characteristics with Hilbert spectrum. The time series of SS, SCS, ECS, and YS, were found to have high energy distributions in the low frequency region (about 12 months per cycle or 0.08 cycles/month), while SJ, OC, KC, and SO showed multiple energy peaks. For MEI, a high energy (variance) concentration was found in the low frequency region (0–0.03 cycles/month) and with a dominant Frequency on 0.0059 cycles/month.

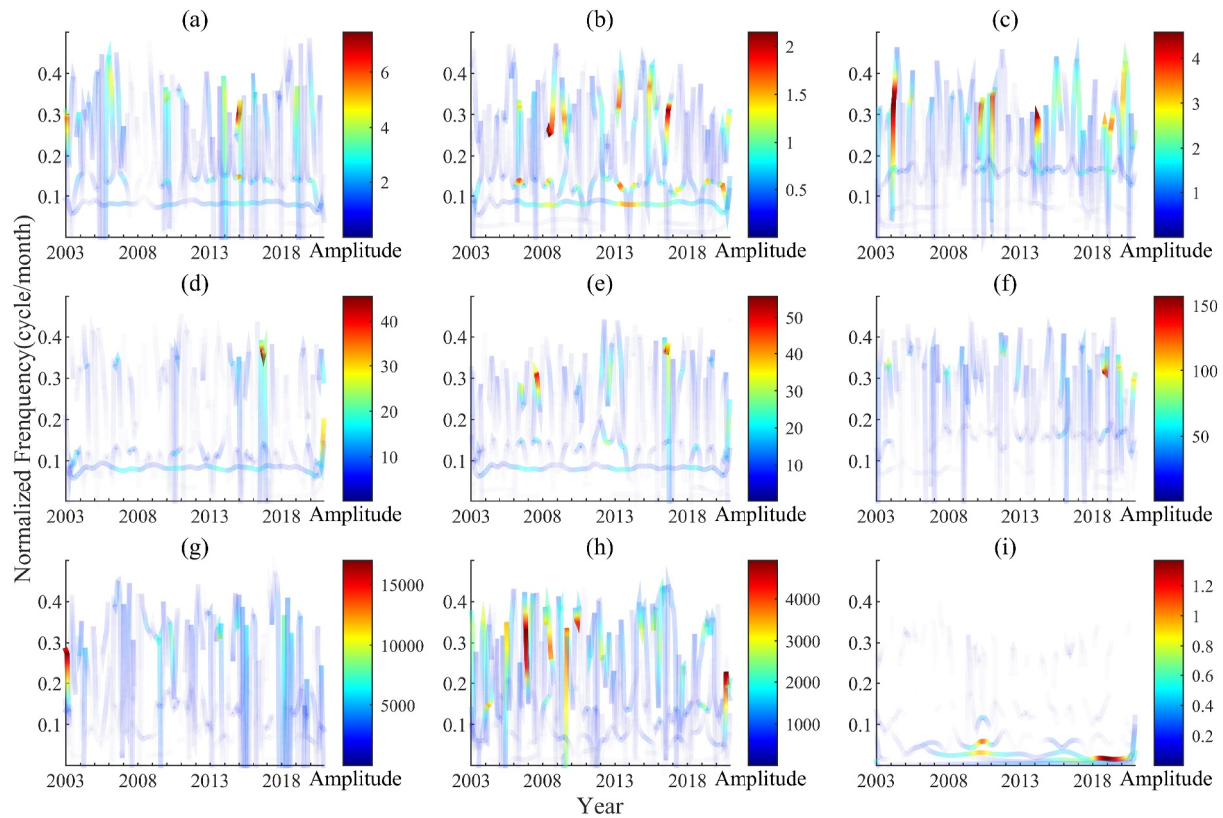


Figure 7. Hilbert spectrum of BI in eight Western North Pacific LMEs - SS (a), SCS (b), KC (c), ECS (d), YS (e), SJ (f), OC (g) and SO (h), and MEI (i).

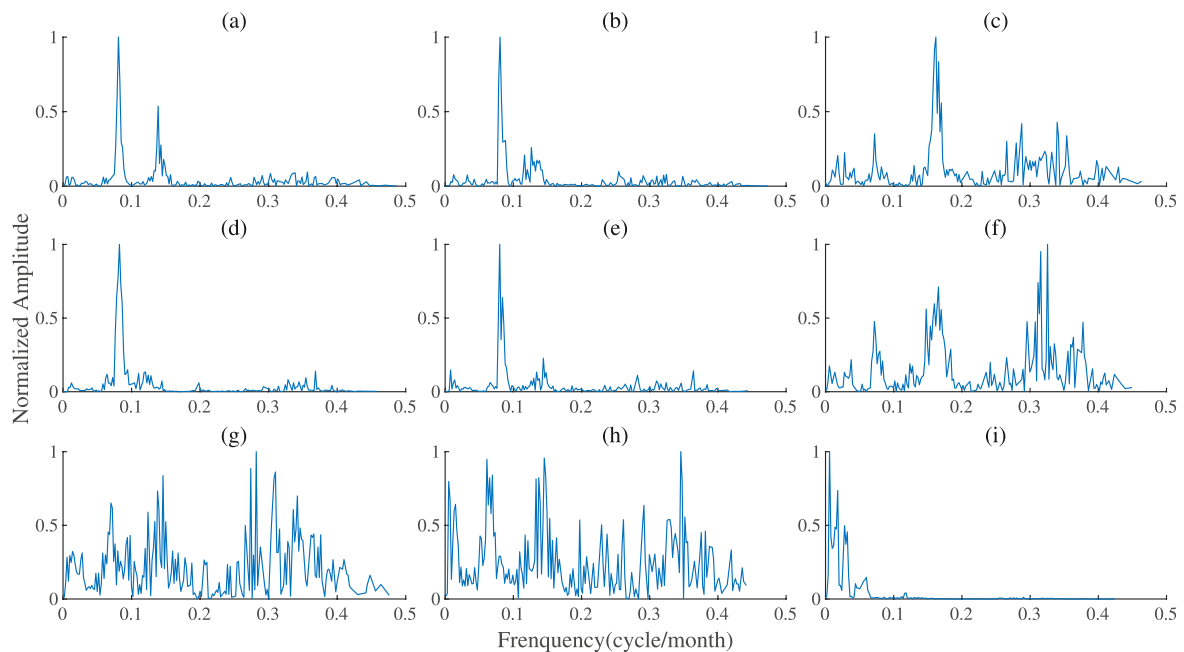


Figure 8. Normalized Hilbert marginal spectrum of BI in eight Western North Pacific LMEs - SS (a), SC S (b), KC (c), ECS (d), YS (e), SJ (f), OC (g) and SO(h), and MEI (i).

Table 2
Statistics of HHT Results of BI in Eight Western North Pacific LMEs and MEI

LMEs	Entropy	Dominant frequency	Energy ratios
SS	5.9875	0.0808	11.07%
SCS	5.9309	0.0805	11.96%
KC	6.5340	0.1618	5.32%
ECS	5.5558	0.0823	11.49%
YS	5.9151	0.0800	12.84%
SJ	6.7218	0.3256	3.72%
OC	7.0748	0.2819	2.39%
SO	7.0498	0.3455	2.11%
MEI	4.8145	0.0059	13.15%

For a quantitative examination of the marginal spectral energy distributions' degree of dispersion across all LMEs, we conducted an analysis involving the computation of marginal spectrum entropy and the energy ratios of dominant frequencies (Table 2). Within the eight LMEs, entropy values were observed to be less than 6, accompanied by a dominant frequency energy ratios exceeding 10% when it comes to SS, SCS, ECS, and YS. Conversely, SJ, OC, KC, and SO exhibited entropy values surpassing 6, while the dominant frequency energy ratios remained below 10%.

3.3.3. Wavelet Coherency Analysis

Wavelet analysis is able to explore the correlation between different signals from a temporal multi-scale perspective. The wavelet coherency spectra showed a robust coherency between the ENSO (MEI) and phytoplankton blooms (BI) across the eight LMEs. The coherency is notably pronounced in the eight LMEs, exhibiting an average periodicity of approximately 57-month in SS, KC, ECS, YS, SJ, and SO, while SCS and OC display an average periodicity of about 24 months (Figure 9 and Table 3). The phase relationships between BI and MEI indicate time lags ranging from $-5 \sim 13$, $-12 \sim 12$, $-4 \sim 10$, $-20 \sim -5$, $-37 \sim 37$, $-22 \sim 22$, $-12 \sim 12$, $-18 \sim 20$ months for the SS, SCS, KC, ECS, YS, SJ, OC, and SO LMEs, respectively. These values are determined based on the range of phase differences and the corresponding periods. Notably, the phase differences between MEI and the BI time series, particularly around the 57-month band, have exhibited increasing coherence in recent years across the SS, KC, ECS, YS, and SJ LMEs. Moreover, these LMEs have demonstrated a cyclical pattern of “large-small-large”

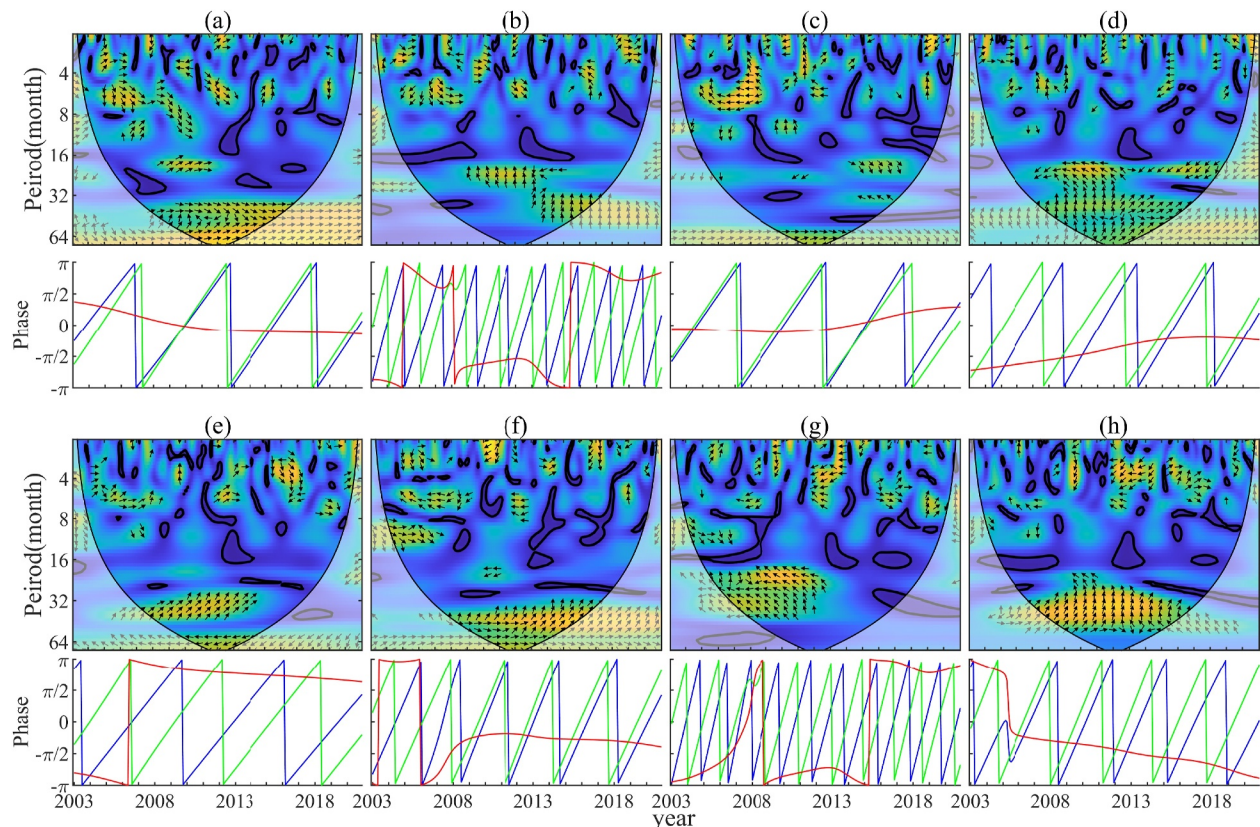


Figure 9. Wavelet coherency and the phase variation between BI in eight Western North Pacific LMEs and MEI. (a)–(h) represent the coherency for SS, SCS, KC, ECS, YS, SJ, OC, and SO. In the wavelet coherency spectrum, the colors represent the coherency values: from blue (0) to yellow (1); black line represents the cone of influence that delimits the region that is not influenced by edge effects; the dash dot line shows a $\alpha = 5\%$ significance level computed based on 500 simple bootstrap. In the phase variation, the green line, blue line, and red line represent the phase of the MEI, the phase of BI, and the phase difference.

Table 3
Statistics of Wavelet Results of BI in Eight Western North Pacific LMEs and MEI

LMEs	Highest coherency	Period (month)	Range of phase difference (π)	Time lag (month)
SS	0.8117	62.40	−0.14–0.37	−5–13
SCS	0.4930	23.38	−1.00–1.00	−12–12
KC	0.6968	66.11	−0.11–0.29	−4–10
ECS	0.6247	55.60	−0.73–−0.20	−20–−5
YS	0.7117	74.21	−1.00–1.00	−37–37
SJ	0.7434	44.13	−1.00–1.00	−22–22
OC	0.4930	23.38	−1.00–1.00	−12–12
SO	0.6507	41.65	−0.89–0.98	−18–20

in the variation of phase differences. In addition, SCS, SJ, and OC all experienced a notable phase mutation around the year 2010.

4. Discussion

4.1. Periodicity and Synchronization of the BI and MEI Time Series in LMEs

Since the relationship between phytoplankton blooms and climate oscillations is rather complicated (Lao et al., 2023; Gao et al., 2023) and it is the first step to conduct a retrospective analysis of past extreme events to gain a better understanding of the commonalities and differences across various systems and their impacts (Michalak, 2016). It is a possible route by employing time-frequency analysis to explore the periodicity and synchronization between them (Xiao et al., 2019). Results confirmed the same significant periodic bands during the same period, and responses of the eight Western North Pacific LMEs were synchronized to some extent.

The EEMD results indicated that the high-frequency periods (IMFs, especially for periods around 3.6 and 12 months' band, Table 1) of LMEs exhibited no differences to those of MEI, showing a significant synchronization of the global climate oscillations and the phytoplankton blooms (Martinez et al., 2009; Watanabe et al., 2005). Hilbert spectra and marginal spectra further showed the similarities of variations in BI and MEI: continuous and high energy/variance fluctuations existed in low-frequency region of all the time series, with a similar variation pattern (Figures 7 and 8). However, the discrete fluctuations existed in the high-frequency region of the BI in all eight LMEs, but not for MEI, showing that ENSO (MEI) cannot fully explain the phytoplankton blooms. Furthermore, the result also clearly reveals the variability of different regions, the eight LMEs in the Western North Pacific can be divided into two main categories: (a). LMEs within the Asia-Europe plate (SS, SCS, ECS, SJ, and YS); (b). LMEs within the Pacific plate (KC, OC, and SO), with the former concentrating their energy in the high-frequency region, which is more in line with the characterization of the marginal spectra of the MEI, and the latter dispersing their energy. This phenomenon suggests that the LMEs within the Asia-Europe Plate are less affected by the circulation within the Pacific Ocean due to the partitioning of the Japanese archipelago, the Sumatra island chain, and the Philippine Arc and that the LMEs within the Asia-Europe Plate are mainly affected by climate teleconnection with ENSO (Yang et al., 2017; Yu & Qu, 2013), and that the climate oscillations influences phytoplankton within the five LMEs by affecting the global climate change. The LMEs within the Pacific plate, on the other hand, are influenced by the oceanic circulation within the Pacific Ocean (Rudnick et al., 2015) and therefore show a more complex dynamic pattern. In addition to this, the greater concentration of energy in the marginal spectra of the five LMEs that are farther away from the ENSO core in terms of distance is also a reflection of the filtering effect of the atmosphere and ocean on the climate oscillations (Donders et al., 2013; Shariff et al., 2015; Zhang et al., 2009). This filtering effect, resulting from coupling of the atmosphere and ocean along with teleconnection, plays a crucial role in regulating and stabilizing the response of marine ecosystems to climate oscillations. This mechanism operates to smooth out responses, rendering corresponding processes relatively gradual and systematic. Consequently, it mitigates abrupt changes, enhances climate system stability, and reduces the extreme impacts of climate change on marine ecosystems. This result suggests two potentially different driving modes of ENSO for phytoplankton blooms in the Western North Pacific. Although the conclusion is only a hypothesis from a data-driven perspective, and the physical mechanism of

the internal drive needs to be further investigated, it is a potential avenue for studying climate teleconnection under complex conditions.

Wavelet approaches can help answer some of the fundamental questions associated with climate variability and phytoplankton blooms, by considering the non-stationary nature of these time series and the possible occurrence of transient states. Our results indicated that phytoplankton BI and ENSO were closely associated with each other, at least statistically, and that these associations appeared transiently in some periods (Figure 9). Wavelet coherence analysis reveals the existence of a cyclic mode between BI and MEI with about 57-month in most LMEs. Previous studies have shown that there is a significant 20.8-year oscillation of CHLa in the surface mixed layer of the Western North Pacific Ocean from 1971 to 2000 (Watanabe et al., 2005) and that jellyfish populations oscillate on a worldwide basis with an approximate 20-year cycle (Condon et al., 2013), reflecting the fact that climate oscillations have an impact on phytoplankton, the primary producer of ecosystems, and further affect the advanced consumers. The 57-month (about 5-year) mode pattern is approximately 1/4 of the 20-year cycle, which not only falls into the oscillation of the 20-year cycle but also shows good resolution in oscillation analysis using advanced time-frequency methods. Furthermore, the sudden phase shift in the year 2010 corresponds well to the sudden shift of ENSO from the El Niño phase to the La Niña phase during this period and the influence of the shift to phytoplankton has been reported by Tian et al. (2021). Besides, the phenomenon that the phase difference tends to be stable in the phase analysis indicates that the connection between MEI and BI has become stronger in recent years, suggesting that the synchronisation between the two is getting stronger. Relevant scholars have demonstrated that global warming accelerates this phenomenon similarly through the carbon cycle feedback between climate and biosphere through coupling of climate models (Cox et al., 2000).

Each of these analytical methods reveals different periodicity and synchronization characteristics between phytoplankton blooms and climate oscillations, which provides an effective theoretical basis for phytoplankton blooms forecasting and early warning as well as material-energy cycling in marine ecosystems on a global scale.

4.2. Multiscale Effect of ENSO on Phytoplankton Blooms

The results of seasonal cycle demonstrate the influence at monthly scales in different LMEs. In tropical regions, a significant negative correlation between SST and BI due to SST closely approaching the optimal temperature threshold for phytoplankton growth. This correlation underscores the localized inhibitory impact of SST rise on BI at the monthly scale. In stark contrast, the peaks of BI curves in LMEs situated in temperate regions precede the peaks of the SST curves by 1–2 months, indicating a lack of promotional effects on BI during the summer months (Liu et al., 2011; Hunter-Cevera et al., 2016). Notably, in LMEs at higher latitudes, the BI exhibits a decreasing trend when SST reaches its peak in summer, suggesting that the optimal temperature conducive to phytoplankton growth in this region is lower. The diverse responses of BI in different LMEs comprehensively illustrate regional variations in the promotion and suppression of phytoplankton by SST, as well as interspecies disparities in algae sensitivity to optimal SST (Hunter-Cevera et al., 2016). When it comes to MLD, the situation is the opposite of SST. In general, MLD affects phytoplankton blooms by controlling nutrient inputs and water stability, and a deeper MLD may lead to a more stable water body and a more adequate supply of nutrients, which would be favorable to the accumulation and development of phytoplankton blooms (Karimpour et al., 2018; Lavigne et al., 2013). However, MLD and BI showed a significant negative correlation in all LMEs in the study area except SS. This suggests that in most LMEs, MLD is no longer the dominant factor limiting phytoplankton blooms, highlighting the importance of SST. Conversely, deeper MLD can actually facilitate phytoplankton blooms due to the more favorable SST conditions in SS. This contrasts with the other LMEs where deeper MLD tends to inhibit phytoplankton blooms, underscoring the unique dynamics of the SS ecosystem. Furthermore, These characteristics mean whether the El Niño phase and La Niña phase of the ENSO influences phytoplankton in the Western North Pacific region through warming or cooling SST, as well as through currents that influence MLD, thus establishing a bridge between climate oscillations and phytoplankton blooms.

The LOESS fitting on annual scales demonstrates the direct correlation between BI and MEI in different LMEs in different years. As the MEI increased, BI showed a trend of increasing then decreasing then increasing then decreasing. This synchronisation of trends over a wide range reflects the intrinsic mechanism by which phytoplankton growth is affected by the climate oscillations, that is, by changes in SST, which affects its biological activity to influence the reproduction of phytoplankton as a primary producer (Doney, 2006). Warmer or colder

SST in both the El Niño phase and La Niña phase lead to a decrease in phytoplankton biomass at the macro level, demonstrating once again the impact of climate extremes on ecosystems.

The correlation between ENSO and the intricate phytoplankton has been elucidated by leveraging SST and MLD as an instrumental bridge. Overall, the time scales of phytoplankton blooms show some similarity, with higher temperatures promoting phytoplankton growth and reproduction to some extent. However, excessively high temperatures can in turn hinder phytoplankton growth and reproduction, a phenomenon that is particularly evident at both high and low latitudes. This phenomenon can be attributed to interspecific differences in the optimal growth temperatures of different phytoplankton species. The identified patterns offer novel insights for comprehensively investigating the evolutionary trajectories of phytoplankton and coastal zone ecosystems on a broad scale in response to the complex interplay of climate change.

4.3. Teleconnection Between the Coastal Phytoplankton Blooms and ENSO

Climate change affects the primary production, that is, phytoplankton, in the aquatic environment (Behrenfeld et al., 2006; Doney, 2006; Doney et al., 2012). It has been confirmed that the phytoplankton was climate-driven, either in basin scale or on a more small scale, like bays, estuaries (Santos et al., 2021, 2022). Nevertheless, much less knowledge is known about the marine scale due to the complex influences, and lack of long-term observations. Our results show that, in terms of overall trends, all of the eight Western North Pacific LMEs exhibit relatively high BI in LMEs during strong El Niño phase and plateau or decrease during strong La Niña phase compared to normal years (Figures 5 and 6). This is particularly evident in the two LMEs, SCS and SS, in the south. The strong El Niño phase has been associated with colder temperature in the Western North Pacific; conversely, strong La Niña phase is associated with warmer temperature (Shabbar & Khandekar, 1996).

Therefore, this cooler oceanic environment in the southern tropical LMEs is more favorable for the growth of common marine phytoplankton, such as cyanobacteria, diatoms, and dinoflagellates, leading to phytoplankton blooms in the strong El Niño years. Furthermore, the BI of other LMEs also shows this trend, which indicates that under the overall trend of global warming, the increase in SST has become a constraint limiting the phytoplankton blooms to the biomass abundance of the whole coastal ecosystem on an interannual scale. This is contrary to the traditional notion that higher temperatures and better light conditions promote phytoplankton growth and higher phytoplankton biomass (Doney, 2006; Gerten & Adrian, 2002). This also reflects the complexity of the impact of climate oscillations on coastal zone ecosystems on large time scales as well as spatial scales.

Given consideration to the patterns shown in this study and previous research, it seems like the climate-driven oscillation of phytoplankton follows certain global principles in different scales. However, compared to the small and medium scales (i.e., bays, estuaries), the cases of marine scales could be much more complicated due to spatial variability in eutrophic levels, natural geographic conditions, and biogeochemical processes. For instance, different intrinsic patterns were found in HHT results: Marginal spectrum energy of MEI was concentrated in the low frequency region (Which is the same in five LMEs), However, the SJ, OC and SO LMEs showed opposite aggregation tendency. This again demonstrates the complexity of coastal ecosystems and the teleconnection between coastal phytoplankton blooms in Western North Pacific and El Niño–Southern Oscillation. Research efforts are needed to further explore the regional specific patterns under the global synchronization of phytoplankton to climate oscillations.

4.4. Potential Influence of Other Environmental Factors

In addition to temperature, precipitation in the Pacific Western North LMEs is also influenced by ENSO, which offers the possibility of establishing a teleconnection between ENSO and phytoplankton blooms in addition to SST. Precipitation may have a significant effect on salinity as well as nutrient in coastal zone areas.

Salinity in LMEs is affected by climate oscillations through temperature and precipitation. Increases in salinity can be due to warm and dry periods inducing high evaporation. Otherwise, a decrease in salinity is generally caused by freshwater inputs from rain, runoff, or floods. When the salinity variations result from saltier water inflow or from freshwater inputs, nutrients may also be input (Fouilland et al., 2012; Pecqueur et al., 2011). Nutrient input from runoff, precipitation, or sediment resuspension is widely considered a key factor triggering blooms in coastal zones. Under conditions favorably warm, an increase in the metabolic demands per unit biomass prompts a heightened need for nutritional supply to sustain the growth of phytoplankton. With escalating

nutrient demands, nutrient stress, and competition intensify, diminishing the benefits accruing to phytoplankton (Daufresne et al., 2009; Hayashida et al., 2020; Peter & Sommer, 2012). This also provides a possible explanation for the inconsistency between the peak of the SST curve and the peak of the BI curve in the mid-latitudes of Figure 3. Specifically, during the process of rising SST, prolific phytoplankton reproduction occurs, potentially leading to nutrient deficiencies during the summer, thereby acting as a possible restraint on further phytoplankton blooms (Song & Yu, 2007). As the role of salinity is not quantitatively analyzed in this study, given their importance, a better explanation of the noise and abrupt changes in the time-frequency domain analyses could be provided by examining how these environmental variables act on phytoplankton under external forcing from the climate oscillations, which can help us to understand the teleconnection between ENSO and phytoplankton blooms.

5. Conclusion

The phytoplankton blooms are intricate, influenced by a multitude of parameters through nonlinear and stochastic relationships. Advanced time-frequency analysis techniques are highly effective in revealing the interactions between phytoplankton blooms and climate oscillations. Despite the inherent complexity spanning various scales, a remarkable level of periodicity and synchronization has been discerned between phytoplankton blooms and the ENSO represented by the MEI in LMEs.

The HHT analysis has revealed two distinct impact patterns of ENSO on phytoplankton blooms in the Western North Pacific. Specifically, the influence of ENSO on phytoplankton blooms within LMEs located within the Sub-European Plate is significantly influenced by coupling of atmosphere and the ocean. In contrast, LMEs within the Pacific Plate experience more direct influences from ocean currents, resulting in a different response. Wavelet analysis has unveiled robust coherence between the MEI and phytoplankton blooms across all LMEs, exhibiting a distinct 57-month periodicity. Furthermore, a detailed examination of the relationship between BI and MEI through LOESS has illuminated the diverse impacts of different ENSO phases, namely El Niño and La Niña, on phytoplankton blooms. In the broader context of global warming, on an interannual scale, strong El Niño phase exhibits a cooling effect on seawater in the Western North Pacific region, which promotes the growth of phytoplankton. Conversely, strong La Niña phase, with their warming effect on seawater in the Western North Pacific, results in the inhibition of phytoplankton growth. In contrast, at the monthly scale, three different patterns of phytoplankton response to ENSO-induced temperature changes within different LMEs arise due to interspecies differences.

Data Availability Statement

The phytoplankton blooms dataset were obtained from <https://doi.org/10.5281/zenodo.7359262> (Dai et al., 2022). The boundaries of LMEs were obtained from <https://www.sciencebase.gov/catalog/item/55c77722e4b08400b1fd8244> (Sherman et al., 2017). The Multivariate ENSO Index (MEI) were downloaded from the NOAA website at <https://psl.noaa.gov/enso/mei/> (NOAA, 2023). The SST was downloaded from CDS at DOI:10.24381/cds.adbb2d47 (Hersbach et al., 2023). The MLD was downloaded from CDS at DOI:10.24381/cds.67e8eeb7 (C3S, 2019).

References

- Ahn, Y.-H., & Shanmugam, P. (2006). Detecting the red tide algal blooms from satellite ocean color observations in optically complex Northeast-Asia Coastal waters. *Remote Sensing of Environment*, 103(4), 419–437. <https://doi.org/10.1016/j.rse.2006.04.007>
- Amin, R., Zhou, J., Gilerson, A., Gross, B., Moshary, F., & Ahmed, S. (2009). Novel optical techniques for detecting and classifying toxic dinoflagellate *Karenia brevis* blooms using satellite imagery. *Optics Express*, 17(11), 9126–9144. <https://doi.org/10.1364/oe.17.009126>
- Amin, S., Hmelo, L., Van Tol, H., Durham, B., Carlson, L., Heal, K., et al. (2015). Interaction and signalling between a cosmopolitan phytoplankton and associated bacteria. *Nature*, 522(7554), 98–101. <https://doi.org/10.1038/nature14488>
- Behrenfeld, M. J., O'Malley, R. T., Siegel, D. A., McClain, C. R., Sarmiento, J. L., Feldman, G. C., et al. (2006). Climate-driven trends in contemporary ocean productivity. *Nature*, 444(7120), 752–755. <https://doi.org/10.1038/nature05317>
- Behrenfeld, M. J., Randerson, J. T., McClain, C. R., Feldman, G. C., Los, S. O., Tucker, C. J., et al. (2001). Biospheric primary production during an ENSO transition. *Science*, 291(5513), 2594–2597. <https://doi.org/10.1126/science.1055071>
- C3S. (2019). Oras5 global ocean reanalysis monthly data from 1958 to present. [dataset]. Copernicus Climate Change Service (C3S) Climate Data Store (CDS). <https://doi.org/10.24381/cds.67e8eeb7>
- Cai, W., Santoso, A., Collins, M., Dewitte, B., Karamperidou, C., Kug, J.-S., et al. (2021). Changing El Niño-Southern Oscillation in a warming climate. *Nature Reviews Earth and Environment*, 2(9), 628–644. <https://doi.org/10.1038/s43017-021-00199-z>
- Carey, C. C., Hanson, P. C., Lathrop, R. C., & St. Amand, A. L. (2016). Using wavelet analyses to examine variability in phytoplankton seasonal succession and annual periodicity. *Journal of Plankton Research*, 38(1), 27–40. <https://doi.org/10.1093/plankt/fbv116>

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- Carstensen, J., Klais, R., & Cloern, J. E. (2015). Phytoplankton blooms in estuarine and coastal waters: Seasonal patterns and key species. *Estuarine, Coastal and Shelf Science*, 162, 98–109. <https://doi.org/10.1016/j.ecss.2015.05.005>
- Carvalho, F., Kohut, J., Oliver, M. J., & Schofield, O. (2017). Defining the ecologically relevant mixed-layer depth for Antarctica's coastal seas. *Geophysical Research Letters*, 44(1), 338–345. <https://doi.org/10.1002/2016gl071205>
- Cazelles, B., Chavez, M., Berteaux, D., Ménard, F., Vik, J. O., Jenouvrier, S., & Stenseth, N. C. (2008). Wavelet analysis of ecological time series. *Oecologia*, 156(2), 287–304. <https://doi.org/10.1007/s00442-008-0993-2>
- Chang, Y.-L., Miyazawa, Y., Oey, L.-Y., Kodaira, T., & Huang, S. (2017). The formation processes of phytoplankton growth and decline in mesoscale eddies in the western north pacific ocean. *Journal of Geophysical Research: Oceans*, 122(5), 4444–4455. <https://doi.org/10.1002/2017jc012722>
- Cheng, Q., Zhong, F., & Wang, P. (2021). Baseflow dynamics and multivariate analysis using bivariate and multiple wavelet coherence in an alpine endorheic river basin (northwest China). *The Science of the Total Environment*, 772, 145013. <https://doi.org/10.1016/j.scitotenv.2021.145013>
- Colijn, F., & Cadée, G. C. (2003). Is phytoplankton growth in the wadden sea light or nitrogen limited? *Journal of Sea Research*, 49(2), 83–93. [https://doi.org/10.1016/s1385-1101\(03\)00002-9](https://doi.org/10.1016/s1385-1101(03)00002-9)
- Condon, R. H., Duarte, C. M., Pitt, K. A., Robinson, K. L., Lucas, C. H., Sutherland, K. R., et al. (2013). Recurrent jellyfish blooms are a consequence of global oscillations. *Proceedings of the National Academy of Sciences*, 110(3), 1000–1005. <https://doi.org/10.1073/pnas.1210920110>
- Cox, P. M., Betts, R. A., Jones, C. D., Spall, S. A., & Totterdell, I. J. (2000). Acceleration of global warming due to carbon-cycle feedbacks in a coupled climate model. *Nature*, 408(6809), 184–187. <https://doi.org/10.1038/35041539>
- Dai, Y., Yang, S., Zhao, D., Hu, C., Xu, W., Anderson, D. M., et al. (2022). Coastal phytoplankton blooms expand and intensify in the 21st century: data and code. [dataset]. *Zenodo*. <https://doi.org/10.5281/zenodo.7359262>
- Dai, Y., Yang, S., Zhao, D., Hu, C., Xu, W., Anderson, D. M., et al. (2023). Coastal phytoplankton blooms expand and intensify in the 21st century. *Nature*, 615(7951), 1–5. <https://doi.org/10.1038/s41586-023-05760-y>
- Daufresne, M., Lengfellner, K., & Sommer, U. (2009). Global warming benefits the small in aquatic ecosystems. *Proceedings of the National Academy of Sciences*, 106(31), 12788–12793. <https://doi.org/10.1073/pnas.0902080106>
- Defrie, E. J., Sheppard, L. W., Reid, P. C., & Reuman, D. C. (2016). Climate change-related regime shifts have altered spatial synchrony of plankton dynamics in the north sea. *Global Change Biology*, 22(6), 2069–2080. <https://doi.org/10.1111/gcb.13229>
- Donders, T. H., Punyasena, S. W., De Boer, H. J., & Wagner-Cremer, F. (2013). ENSO signature in botanical proxy time series extends terrestrial el Niño record into the (sub) tropics. *Geophysical Research Letters*, 40(21), 5776–5781. <https://doi.org/10.1002/2013gl018038>
- Doney, S. C. (2006). Plankton in a warmer world. *Nature*, 444(7120), 695–696. <https://doi.org/10.1038/444695a>
- Doney, S. C., Ruckelshaus, M., Emmett Duffy, J., Barry, J. P., Chan, F., English, C. A., et al. (2012). Climate change impacts on marine ecosystems. *Annual Review of Marine Science*, 4(1), 11–37. <https://doi.org/10.1146/annurev-marine-041911-111611>
- Fouilland, E., Trottet, A., Bancon-Montigny, C., Bouvy, M., Le Floch, E., Gonzalez, J.-L., et al. (2012). Impact of a river flash flood on microbial carbon and nitrogen production in a mediterranean lagoon (thau lagoon, France). *Estuarine, Coastal and Shelf Science*, 113, 192–204. <https://doi.org/10.1016/j.ecss.2012.08.004>
- Gao, Y., Yang, L., Liu, H., & Xie, Z. (2023). Positive atlantic multidecadal oscillation has driven poleward redistribution of the west antarctic peninsula biota through a. *The Science of the Total Environment*, 881, 163373. <https://doi.org/10.1016/j.scitotenv.2023.163373>
- Gerten, D., & Adrian, R. (2002). Effects of climate warming, north Atlantic oscillation, and El Niño-Southern Oscillation on thermal conditions and plankton dynamics in northern hemispheric lakes. *TheScientificWorldJOURNAL*, 2, 586–606. <https://doi.org/10.1100/tsw.2002.141>
- Glé, C., Del Amo, Y., Bec, B., Sautour, B., Froidefond, J.-M., Gohin, F., et al. (2007). Typology of environmental conditions at the onset of winter phytoplankton blooms in a shallow macrotidal coastal ecosystem, arcachon bay (France). *Journal of Plankton Research*, 29(11), 999–1014. <https://doi.org/10.1093/plankt/fbm074>
- Gordon, H. R., Clark, D. K., Mueller, J. L., & Hovis, W. A. (1980). Phytoplankton pigments from the nimbus-7 coastal zone color scanner: Comparisons with surface measurements. *Science*, 210(4465), 63–66. <https://doi.org/10.1126/science.210.4465.63>
- Grinsted, A., Moore, J. C., & Jevrejeva, S. (2004). Application of the cross wavelet transform and wavelet coherence to geophysical time series. *Nonlinear Processes in Geophysics*, 11(5/6), 561–566. <https://doi.org/10.5194/npg-11-561-2004>
- Guo, X., Gao, Y., Zhang, S., Wu, L., Chang, P., Cai, W., et al. (2022). Threat by marine heatwaves to adaptive large marine ecosystems in an eddy-resolving model. *Nature Climate Change*, 12(2), 179–186. <https://doi.org/10.1038/s41558-021-01266-5>
- Gutierrez, R. G., Linhart, J. M., & Pitblado, J. S. (2003). From the help desk: Local polynomial regression and Stata plugins. *STATA Journal*, 3(4), 412–419. <https://doi.org/10.1177/1536867x0300300409>
- Hayashida, H., Matear, R. J., & Strutton, P. G. (2020). Background nutrient concentration determines phytoplankton bloom response to marine heatwaves. *Global Change Biology*, 26(9), 4800–4811. <https://doi.org/10.1111/gcb.15255>
- Hersbach, H., Bell, B., Berrisford, P., Biavati, G., Horányi, A., Muñoz Sabater, J., et al. (2023). ERA5 hourly data on single levels from 1940 to present. [Dataset]. *Copernicus Climate Change Service (C3S) Climate Data Store (CDS)*. <https://doi.org/10.24381/cds.adbb2d47>
- Hieronymus, J., Eilola, K., Hieronymus, M., Meier, H., Saraiva, S., & Karlson, B. (2018). Causes of simulated long-term changes in phytoplankton biomass in the baltic proper: A wavelet analysis. *Biogeosciences*, 15(16), 5113–5129. <https://doi.org/10.5194/bg-15-5113-2018>
- Hou, X., Feng, L., Dai, Y., Hu, C., Gibson, L., Tang, J., et al. (2022). Global mapping reveals increase in lacustrine algal blooms over the past decade. *Nature Geoscience*, 15(2), 130–134. <https://doi.org/10.1038/s41561-021-00887-x>
- Hu, C., & Feng, L. (2017). Modified modis fluorescence line height data product to improve image interpretation for red tide monitoring in the eastern gulf of Mexico. *Journal of Applied Remote Sensing*, 11(1), 012003. <https://doi.org/10.1177/1.jrs.11.012003>
- Hu, C., Feng, L., Lee, Z., Franz, B. A., Bailey, S. W., Werdell, P. J., & Proctor, C. W. (2019). Improving satellite global chlorophyll a data products through algorithm refinement and data recovery. *Journal of Geophysical Research: Oceans*, 124(3), 1524–1543. <https://doi.org/10.1029/2019jc014941>
- Hu, C., Lee, Z., & Franz, B. (2012). Chlorophyll algorithms for oligotrophic oceans: A novel approach based on three-band reflectance difference. *Journal of Geophysical Research: Oceans*, 117(C1). <https://doi.org/10.1029/2011jc007395>
- Hu, C., Muller-Karger, F. E., Taylor, C. J., Carder, K. L., Kelble, C., Johns, E., & Heil, C. A. (2005). Red tide detection and tracing using modis fluorescence data: A regional example in SW Florida coastal waters. *Remote Sensing of Environment*, 97(3), 311–321. <https://doi.org/10.1016/j.rse.2005.05.013>
- Hu, J., Wang, X., & Qin, H. (2016). Novel and efficient computation of Hilbert–Huang transform on surfaces. *Computer Aided Geometric Design*, 43, 95–108. <https://doi.org/10.1016/j.cagd.2016.02.011>
- Huang, B., Xiang, W., Zeng, X., Chiang, K.-P., Tian, H., Hu, J., et al. (2011). Phytoplankton growth and microzooplankton grazing in a subtropical coastal upwelling system in the Taiwan Strait. *Continental Shelf Research*, 31(6), S48–S56. <https://doi.org/10.1016/j.csr.2011.02.005>

- Huang, N. E., Shen, Z., Long, S. R., Wu, M. C., Shih, H. H., Zheng, Q., et al. (1998). The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis. *Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences*, 454(1971), 903–995. <https://doi.org/10.1098/rspa.1998.0193>
- Huang, N. E., & Wu, Z. (2008). A review on Hilbert-Huang transform: Method and its applications to geophysical studies. *Reviews of Geophysics*, 46(2). <https://doi.org/10.1029/2007rg000228>
- Hunter-Cevera, K. R., Neubert, M. G., Olson, R. J., Solow, A. R., Shalapyonok, A., & Sosik, H. M. (2016). Physiological and ecological drivers of early spring blooms of a coastal phytoplankton. *Science*, 354(6310), 326–329. <https://doi.org/10.1126/science.aaf8536>
- Jerin, J. N., Islam, H. T., Islam, A. R. M. T., Shahid, S., Hu, Z., Badhan, M. A., et al. (2021). Spatiotemporal trends in reference evapotranspiration and its driving factors in Bangladesh. *Theoretical and Applied Climatology*, 144(1–2), 793–808. <https://doi.org/10.1007/s00704-021-03566-4>
- Joh, Y., & Di Lorenzo, E. (2017). Increasing coupling between NPGO and PDO leads to prolonged marine heatwaves in the northeast Pacific. *Geophysical Research Letters*, 44(22), 11–663. <https://doi.org/10.1002/2017gl075930>
- Karimpour, F., Tandon, A., & Mahadevan, A. (2018). Sustainance of phytoplankton in the subpolar north Atlantic during winter. *Journal of Geophysical Research: Oceans*, 123(9), 6531–6548. <https://doi.org/10.1029/2017jc013639>
- Lao, Q., Liu, S., Ling, Z., Jin, G., Chen, F., Chen, C., & Zhu, Q. (2023). External dynamic mechanisms controlling the periodic offshore blooms in Beibu Gulf. *Journal of Geophysical Research: Oceans*, 128(6), e2023JC019689. <https://doi.org/10.1029/2023jc019689>
- Lavigne, H., d'Ortenzio, F., Migon, C., Claustre, H., Testor, P., d'Alcalá, M. R., et al. (2013). Enhancing the comprehension of mixed layer depth control on the Mediterranean phytoplankton phenology. *Journal of Geophysical Research: Oceans*, 118(7), 3416–3430. <https://doi.org/10.1002/jgrc.20251>
- Le, C., Wu, M., Sun, H., Long, S.-m., & Beck, M. W. (2022). Linking phytoplankton variability to atmospheric blocking in an eastern boundary upwelling system. *Journal of Geophysical Research: Oceans*, 127(6), e2021JC017348. <https://doi.org/10.1029/2021jc017348>
- Li, D., Chen, Y., Qi, J., Zhu, Y., Lu, C., & Yin, B. (2023). Attribution of the July 2021 record-breaking northwest Pacific marine heatwave to global warming, atmospheric circulation, and ENSO. *Bulletin of the American Meteorological Society*, 104(1), E291–E297. <https://doi.org/10.1175/bams-d-22-0142.1>
- Li, T., Wang, B., Wu, B., Zhou, T., Chang, C.-P., & Zhang, R. (2017). Theories on formation of an anomalous anticyclone in western North Pacific during El Niño: A review. *Journal of Meteorological Research*, 31(6), 987–1006. <https://doi.org/10.1007/s13351-017-7147-6>
- Li, Y., Yang, D., Xu, L., Gao, G., He, Z., Cui, X., et al. (2022). Three types of typhoon-induced upwellings enhance coastal algal blooms: A case study. *Journal of Geophysical Research: Oceans*, 127(5), e2022JC018448. <https://doi.org/10.1029/2022jc018448>
- Lim, H.-G., Dunne, J. P., Stock, C. A., & Kwon, M. (2022). Attribution and predictability of climate-driven variability in global ocean color. *Journal of Geophysical Research: Oceans*, 127(10), e2022JC019121. <https://doi.org/10.1029/2022jc019121>
- Lin, I., Liu, W. T., Wu, C.-C., Wong, G. T., Hu, C., Chen, Z., et al. (2003). New evidence for enhanced ocean primary production triggered by tropical cyclone. *Geophysical Research Letters*, 30(13). <https://doi.org/10.1029/2003gl017141>
- Liu, X., Lu, X., & Chen, Y. (2011). The effects of temperature and nutrient ratios on microcystis blooms in Lake Taihu, China: An 11-year investigation. *Harmful Algae*, 10(3), 337–343. <https://doi.org/10.1016/j.hal.2010.12.002>
- Liu, Z., Wang, Z., Wang, J., Zhang, Z., Li, D., Yu, Z., et al. (2023). An improved method of the globally resolved energy balance model by the Bayesian networks. *Geoscientific Model Development*, 16(10), 2939–2955. <https://doi.org/10.5194/gmd-16-2939-2023>
- Lu, W., Yan, X.-H., & Jiang, Y. (2015). Winter bloom and associated upwelling northwest of the Luzon Island: A coupled physical-biological modeling approach. *Journal of Geophysical Research: Oceans*, 120(1), 533–546. <https://doi.org/10.1002/2014jc010218>
- Martínez, E., Antoine, D., D'Ortenzio, F., & Gentili, B. (2009). Climate-driven basin-scale decadal oscillations of oceanic phytoplankton. *Science*, 326(5957), 1253–1256. <https://doi.org/10.1126/science.1177012>
- Menge, B. A., Chan, F., Nielsen, K. J., Lorenzo, E. D., & Lubchenco, J. (2009). Climatic variation alters supply-side ecology: Impact of climate patterns on phytoplankton and mussel recruitment. *Ecological Monographs*, 79(3), 379–395. <https://doi.org/10.1890/08-2086.1>
- Michalak, A. M. (2016). Study role of climate change in extreme threats to water quality. *Nature*, 535(7612), 349–350. <https://doi.org/10.1038/535349a>
- Moore, J. K., & Abbott, M. R. (2000). Phytoplankton chlorophyll distributions and primary production in the southern ocean. *Journal of Geophysical Research: Oceans*, 105(C12), 28709–28722. <https://doi.org/10.1029/1999jc000043>
- NOAA. (2023). Multivariate ENSO Index Version 2 (MEI.v2). [dataset]. NOAA Physical Sciences Laboratory. Retrieved from <https://psl.noaa.gov/enso/mei/>
- Paerl, H. W., Hall, N. S., & Calandrino, E. S. (2011). Controlling harmful cyanobacterial blooms in a world experiencing anthropogenic and climatic-induced change. *The Science of the Total Environment*, 409(10), 1739–1745. <https://doi.org/10.1016/j.scitotenv.2011.02.001>
- Paerl, H. W., & Huisman, J. (2008). Blooms like it hot. *Science*, 320(5872), 57–58. <https://doi.org/10.1126/science.1155398>
- Paerl, H. W., & Paul, V. J. (2012). Climate change: Links to global expansion of harmful cyanobacteria. *Water Research*, 46(5), 1349–1363. <https://doi.org/10.1016/j.watres.2011.08.002>
- Palacz, A. P., Xue, H., Armbricht, C., Zhang, C., & Chai, F. (2011). Seasonal and inter-annual changes in the surface chlorophyll of the South China Sea. *Journal of Geophysical Research: Oceans*, 116(C9), C09015. <https://doi.org/10.1029/2011jc007064>
- Pecqueur, D., Vidussi, F., Fouilland, E., Le Floch, E., Mas, S., Roques, C., et al. (2011). Dynamics of microbial planktonic food web components during a river flash flood in a Mediterranean coastal lagoon. *Hydrobiologia*, 673(1), 13–27. <https://doi.org/10.1007/s10750-011-0745-x>
- Peter, K. H., & Sommer, U. (2012). Phytoplankton cell size: Intra- and interspecific effects of warming and grazing. *PLoS One*, 7(11), e49632. <https://doi.org/10.1371/journal.pone.0049632>
- Power, S., Lengaigne, M., Capotondi, A., Khodri, M., Vialard, J., Jebri, B., et al. (2021). Decadal climate variability in the tropical Pacific: Characteristics, causes, predictability, and prospects. *Science*, 374(6563), eaay9165. <https://doi.org/10.1126/science.aay9165>
- Rousseaux, C. S., & Gregg, W. W. (2013). Interannual variation in phytoplankton primary production at a global scale. *Remote Sensing*, 6(1), 1–19. <https://doi.org/10.3390/rs6010001>
- Rudnick, D. L., Jan, S., & Lee, C. M. (2015). A new look at circulation in the western North Pacific: Introduction to the special issue. *Oceanography*, 28(4), 16–23. <https://doi.org/10.5670/oceanog.2015.77>
- Santos, M., Amorim, A., Brotas, V., Cruz, J., Palma, C., Borges, C., et al. (2022). Spatio-temporal dynamics of phytoplankton community in a well-mixed temperate estuary (Sado Estuary, Portugal). *Scientific Reports*, 12(1), 16423. <https://doi.org/10.1038/s41598-022-20792-6>
- Santos, M., Moita, M. T., Oliveira, P. B., & Amorim, A. (2021). Phytoplankton communities in two wide-open bays in the Iberian upwelling system. *Journal of Sea Research*, 167, 101982. <https://doi.org/10.1016/j.seares.2020.101982>
- Schulien, J. A., Behrenfeld, M. J., Hair, J. W., Hostetler, C. A., & Twardowski, M. S. (2017). Vertically-resolved phytoplankton carbon and net primary production from a high spectral resolution lidar. *Optics Express*, 25(12), 13577–13587. <https://doi.org/10.1364/oe.25.013577>
- Shabbar, A., & Khandekar, M. (1996). The impact of El Niño–Southern Oscillation on the temperature field over Canada: Research note. *Atmosphere-Ocean*, 34(2), 401–416. <https://doi.org/10.1080/07055900.1996.9649570>

- Shariff, A. R. M., Singh, M. S. J., Chellappan, K., Suparta, W., Tangang, F. T., Salimun, E., et al. (2015). Initial observations of cold surge frequency over southeast asia in relation to ENSO-induced anomalies. In *2015 International Conference on Space Science and Communication (Iconspace)* (pp. 453–458).
- Sherman, K. L. R., Shuford, Adams, S., & Bosch, J. (2017). Large marine ecosystems. [dataset]. *USGS ScienceBase*. Retrieved from <https://psl.noaa.gov/enso/mei/>
- Shibano, R., Yamanaka, Y., Okada, N., Chuda, T., Suzuki, S.-i., Niino, H., & Toratani, M. (2011). Responses of marine ecosystem to typhoon passages in the western subtropical north Pacific. *Geophysical Research Letters*, 38(18). <https://doi.org/10.1029/2011gl048717>
- Somu, N., Mr, G. R., & Ramamritham, K. (2021). A deep learning framework for building energy consumption forecast. *Renewable and Sustainable Energy Reviews*, 137, 110591. <https://doi.org/10.1016/j.rser.2020.110591>
- Song, X., & Yu, Z. (2007). Nutrient effect on phytoplankton in typical mariculture waters in summer in the northeast of the Jiaozhou bay. *Oceanologia et Limnologia Sinica*, 38(5), 452.
- Sun, J.-Z., Wang, T., Huang, R., Yi, X., Zhang, D., Beardall, J., et al. (2022). Enhancement of diatom growth and phytoplankton productivity with reduced O₂ availability is moderated by rising CO₂. *Communications Biology*, 5(1), 54. <https://doi.org/10.1038/s42003-022-03006-7>
- Temmerman, S., Meire, P., Bouma, T. J., Herman, P. M., Ysebaert, T., & De Vriend, H. J. (2013). Ecosystem-based coastal defence in the face of global change. *Nature*, 504(7478), 79–83. <https://doi.org/10.1038/nature12859>
- Tian, F., Zhang, R.-H., & Wang, X. (2021). Indian ocean warming as a potential trigger for super phytoplankton blooms in the eastern equatorial Pacific from El Niño to La Niña transitions. *Environmental Research Letters*, 16(5), 054040. <https://doi.org/10.1088/1748-9326/abf76f>
- Tian, T., Su, J., Flöser, G., Wiltshire, K., & Wirtz, K. (2011). Factors controlling the onset of spring blooms in the German bight 2002–2005: Light, wind and stratification. *Continental Shelf Research*, 31(10), 1140–1148. <https://doi.org/10.1016/j.csr.2011.04.008>
- Trombetta, T., Vidussi, F., Mas, S., Parin, D., Simier, M., & Mostajir, B. (2019). Water temperature drives phytoplankton blooms in coastal waters. *PLoS One*, 14(4), e0214933. <https://doi.org/10.1371/journal.pone.0214933>
- Tiyasha, T., Tung, T. M., & Yaseen, Z. M. (2020). A survey on river water quality modelling using artificial intelligence models: 2000–2020. *Journal of Hydrology*, 585, 124670. <https://doi.org/10.1016/j.jhydrol.2020.124670>
- Vidya, P., Balaji, M., & Murali, R. M. (2021). Cyclone Hudhud-Eddy induced phytoplankton bloom in the northern bay of Bengal using a coupled model. *Progress in Oceanography*, 197, 102631. <https://doi.org/10.1016/j.pocean.2021.102631>
- Waga, H., Hirawake, T., & Ueno, H. (2019). Impacts of mesoscale eddies on phytoplankton size structure. *Geophysical Research Letters*, 46(22), 13191–13198. <https://doi.org/10.1029/2019gl085150>
- Waltham, N. J., Elliott, M., Lee, S. Y., Lovelock, C., Duarte, C. M., Buelow, C., et al. (2020). Un decade on ecosystem restoration 2021–2030—What chance for success in restoring coastal ecosystems? *Frontiers in Marine Science*, 7, 71. <https://doi.org/10.3389/fmars.2020.00071>
- Watanabe, Y. W., Ishida, H., Nakano, T., & Nagai, N. (2005). Spatiotemporal decreases of nutrients and chlorophyll-a in the surface mixed layer of the western north Pacific from 1971 to 2000. *Journal of Oceanography*, 61(6), 1011–1016. <https://doi.org/10.1007/s10872-006-0017-y>
- Watorek, M., Drozd, S., Kwapien, J., Minati, L., Oswiecimka, P., & Stanuszek, M. (2021). Multiscale characteristics of the emerging global cryptocurrency market. *Physics Reports*, 901, 1–82. <https://doi.org/10.1016/j.physrep.2020.10.005>
- Wolter, K., & Timlin, M. S. (2011). El Niño/Southern Oscillation behaviour since 1871 as diagnosed in an extended multivariate ENSO index (mei-ext). *International Journal of Climatology*, 31(7), 1074–1087. <https://doi.org/10.1002/joc.2336>
- Wu, Z., & Huang, N. E. (2009). Ensemble empirical mode decomposition: A noise-assisted data analysis method. *Advances in Adaptive Data Analysis*, 1(01), 1–41. <https://doi.org/10.1142/s1793536909000047>
- Xiao, X., He, J., Yu, Y., Cazelles, B., Li, M., Jiang, Q., & Xu, C. (2019). Teleconnection between phytoplankton dynamics in north temperate lakes and global climatic oscillation by time-frequency analysis. *Water Research*, 154, 267–276. <https://doi.org/10.1016/j.watres.2019.01.056>
- Yang, H., Wu, L., Sun, S., & Chen, Z. (2017). Role of the south China Sea in regulating the north Pacific double-gyre system. *Journal of Physical Oceanography*, 47(7), 1617–1635. <https://doi.org/10.1175/jpo-d-16-0272.1>
- Yu, K., & Qu, T. (2013). Imprint of the Pacific decadal oscillation on the south China Sea throughflow variability. *Journal of Climate*, 26(24), 9797–9805. <https://doi.org/10.1175/jcli-d-12-00785.1>
- Zhang, L., Chang, P., & Tippet, M. K. (2009). Linking the Pacific meridional mode to ENSO: Utilization of a noise filter. *Journal of Climate*, 22(4), 905–922. <https://doi.org/10.1175/2008jcli2474.1>
- Zhang, L., & Zhou, T. (2015). Drought over east Asia: A review. *Journal of Climate*, 28(8), 3375–3399. <https://doi.org/10.1175/jcli-d-14-00259.1>